

Dynamic ISF from total daily dose — evaluation methodology

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Audit update, 16 September 2026: V2 is the author-confirmed no-+1 equation. The Python replay used that form. The delta-IOB and action-balance estimators are outcome proxies rather than independent physiological ISF measurements. Conclusions relying on them as ground truth are withdrawn. See DYNAMIC-ISF-AUDIT-2026-09.md.

Abstract

This paper describes a reproducible computational method for evaluating equations that set a dynamic insulin sensitivity factor (ISF) as a function of total daily dose (TDD). The method (1) implements each candidate equation exactly, with a unit-test contract; (2) reconstructs, from each person's available insulin-delivery history, the closest available TDD signal to the one the equations consume; (3) replays every glucose reading through each equation to generate the ISF it would produce; (4) compares the resulting ISF distributions between equations; (5) searches a family of candidate equations against a delta-IOB outcome proxy using leave-one-user-out cross-validation; and (6) validates the equation implementations against ISF values that devices themselves logged. We applied it to 171 people using open-source automated insulin delivery (AID) systems and roughly 9 million glucose readings. This paper documents each step, the script that performs it, and the reasoning that connects the outputs to the conclusions.

1. Background and objective

Dynamic ISF makes a person's correction sensitivity a function of two things: their total daily dose of insulin, and their current glucose. The original formulation, due to Chris Wilson, sets the sensitivity anchor in inverse proportion to TDD (the classic "1800 rule" shape) and then scales it logarithmically with glucose. We refer to this as **v1**. A later revision of the maths, also by Chris Wilson, makes the anchor inversely proportional to **TDD squared**. We refer to this as **v2**.

Both forms share the same TDD-blending step but differ on two counts: how strongly TDD drives the sensitivity anchor ($1/\text{TDD}$ against $1/\text{TDD}^2$), and the glucose scaler, where v1 keeps a +1 in its log and v2 does not. The questions this method answers:

1. How differently do v1 and v2 dose, across a real population, as a function of TDD?
2. How does either TDD power law compare with entered ISF and a delta-IOB outcome proxy?

3. Is there a better-fitting equation among a wider candidate family?

The focus is the between-person sensitivity anchor as a function of TDD, since that is where the equations diverge most. The glucose-scaler difference is carried through the replay rather than assumed away. This is a retrospective, decision-level analysis of the equations, not a closed-loop outcome study, and it provides no dosing advice.

2. The equations

Both v1 and v2 first derive a blended TDD, then a sensitivity anchor at "normal target", then scale it by current glucose.

TDD blend (shared). From five windows of recorded delivery:

$W8H = (1.4 \cdot TDD_{4h} + 0.6 \cdot TDD_{8-4h}) \cdot 3$
if $W8H < 0.75 \cdot TDD_{7d}$:
 $adj7D = W8H + (W8H/TDD_{7d}) \cdot (TDD_{7d} - W8H)$
 $TDD = 0.34 \cdot adj7D + 0.33 \cdot TDD_{1d} + 0.33 \cdot W8H$
else:
 $TDD = 0.33 \cdot W8H + 0.34 \cdot TDD_{7d} + 0.33 \cdot TDD_{1d}$

Glucose scaling. With glucose capped at 210 mg/dL (excess at one-third weight), divisor 75 (the standard rapid-acting configuration, including NovoRapid), normal target 99 mg/dL. v1 keeps the +1 in the log; v2 uses $\ln(BG/divisor)$ (no +1) and floors glucose at divisor+1:

v1 scaler: $\ln(target/divisor + 1) / \ln(bg_capped/divisor + 1)$
v2 scaler: $\ln(target/divisor) / \ln(bg_floored/divisor)$ $bg_floored = \max(bg_capped, 76)$

Sensitivity anchor:

	anchor at normal target	implied TDD law
v1	$1800 / (TDD \cdot \ln(target/divisor + 1))$	$ISF \propto 1/TDD$
v2	$2300 / (\ln(target/divisor) \cdot TDD^2 \cdot 0.02)$	$ISF \propto 1/TDD^2$

The ISF used at any glucose is anchor · scaler. In long form:

v1: $ISF(BG) = 1800 / (TDD \cdot \ln(bg_capped/75 + 1))$
v2: $ISF(BG) = 115\,000 / (TDD^2 \cdot \ln(bg_floored/75))$

Because v1 uses $\ln(BG/divisor+1)$ and v2 uses $\ln(BG/divisor)$, the glucose terms differ, so the ratio between the two equations changes with glucose as well as TDD:

$ISF_{v2} / ISF_{v1} \propto (1 / TDD) \cdot \ln(BG/75 + 1) / \ln(BG_floored/75)$

That ratio grows large at low glucose, where the v2 log approaches its floor, and falls toward a small margin when glucose is high. The method therefore replays both axes, reconstructed TDD and glucose, for every reading rather than relying on a single closed form.

3. Data

Three anonymised cohorts of open-source AID users, held in a local time-series database:

Platform	Users	Readings	TDD source
Trio	23	3.03 M	device-logged TDD
AAPS	39	1.31 M	reconstructed from treatments
OpenAPS	109	6.58 M	reconstructed from treatments

Each row is one 5-minute decision cycle carrying the glucose reading, the hour of day, and an anonymised relative timestamp (seconds since that person's first record). Raw delivery and profile archives (boluses, temp basals, basal schedules) remain on disk for the TDD reconstruction. No participant-level data is published; only cohort-level aggregates and figures are released.

4. Method, step by step

The pipeline is a Python package run in three stages plus two analyses. It is parallelised per person (one person = one task), resumable, with atomic per-person outputs and timestamped run manifests, and completes end to end in under ten minutes on a desktop workstation.

4.1 Equation implementation and its test contract

The v1 and v2 equations and the TDD blend are implemented as vectorised array operations, preserving every constant, the glucose cap, and the blend's branch logic. Correctness is fixed by 27 unit tests against hand-computed fixtures: the v2 glucose floor, the glucose-dependent v1/v2 ratio, the glucose-cap compression, both branches of the TDD blend, and the missing-data gates. Every downstream conclusion rests on these being exactly the equations as defined, and the tests are the contract that guarantees it.

4.2 TDD reconstruction

Where the device did not log TDD, we rebuild the five windows the blend consumes from raw delivery records:

1. **Parse delivery events:** boluses (any record carrying insulin) and temp-basal segments (absolute rate or percent-of-profile, with supersession and cancellation), from each person's archive. Where an export has no temp-basal records, basal delivery uses the person's profile schedule (flagged).
2. **Build a 5-minute delivery grid:** lay the basal schedule across time, overlay temp-basal segments, and add boluses into their bins — units delivered per bin.

3. **Compute the windows:** trailing 4 h, the 8–4 h window, trailing 24 h, yesterday's calendar total, and the 7-day average. A calendar day counts toward the 7-day average only if it is complete on the grid and carries at least one bolus, so upload gaps do not silently deflate it.
4. **Blend:** apply the exact W8H formula; emit a missing value (the real-world fallback to profile ISF) wherever a component is absent.

This matters because the equations are functions of the blended TDD, not a flat daily average. Reconstructing the windowed signal makes the replay substantially closer to the device calculation than a flat daily average. It remains approximate where temporary-basal delivery is absent from an export, particularly in the AAPS cohort. A flat-TDD arm runs alongside it to measure how much the blend itself contributes.

4.3 Absolute-time recovery and ISF replay

Reconstructed TDD lives in absolute time; decision rows carry only relative time. To join them we recover each person's absolute-time anchor and **validate it against an independent signal — the recorded hour-of-day**. Two candidate anchors are each refined by a whole-day shift and then a fine sweep, scored by how well the implied hour-of-day matches the record; the candidate giving the best join coverage wins, and residual mismatch is flagged when it exceeds tolerance.

Then, for every glucose reading, the replay computes the ISF under both equations using that reading's TDD (device-logged where available, reconstructed otherwise). The output is a per-person table of (glucose, TDD, anchor, ISF) under each equation. Median join coverage was 99.4%, and 5 of 148 reconstructed users retain an uncertain anchor. As a cross-check, the analytic v2/v1 ratio computed from a reading's own glucose and TDD matches the ratio of the replayed ISFs at sampled readings, which confirms the join is sound.

4.4 Figures

Each person gets a page: the ISF–glucose curves under each equation at their median TDD (with a TDD-interquartile band), a two-week sample of both dynamic-ISF traces over their real glucose, and the per-reading ratio distribution. Cohort figures: the per-person median v2/v1 ratio against TDD; per-person median ISF under each equation; and the log-log ISF–TDD relationship with the observed points and the fitted slope.

4.5 Best-fit equation search

To ask whether a better equation fits these data, candidates are scored against two targets:

- **delta-IOB sensitivity proxy** — a regression on fasting windows ($\Delta BG = a + b \cdot \Delta IOB + c \cdot BG_trend$; proxy = $-b$; $n = 114$ after quality gates). New insulin delivery changes IOB during the window, and delivery responds to glucose. This target does not identify the physiological effect of insulin.
- **tuned-profile ISF** — each person's own profile setting ($n = 138$): "what experienced users converge to".

Candidates: the two equations above; the historical 1700-rule; the tuned profile value; and fitted forms — a re-fitted constant/TDD, a free-exponent power law $A \cdot \text{TDD}^b$, a fixed-exponent square-root rule $K/\sqrt{\text{TDD}}$, TDD-quartile bands, multivariate log-linear models adding carb ratio / basal fraction / target, and a geometric blend of the profile value with the power law.

Every fitted candidate is scored by leave-one-user-out cross-validation — refit on all but one person, predict the held-out person, repeat — so fitted forms are judged out-of-sample and are directly comparable with the fixed rules. Metrics: median absolute error, median absolute log-error (scale-free), and fraction of people predicted within $\pm 30\%$. Without cross-validation the fitted equations would carry an unfair in-sample advantage; holding out each person in turn measures how well they generalise to someone new.

4.6 Implementation validation against device-logged ISF

Independently of the equation comparison, we check that the v1 implementation reproduces what devices actually computed. Trio logs its own per-cycle ISF, giving a device reference for its dynamic-ISF users. We correlate replayed v1 ISF against the device value, expecting, per person, a positive log-log correlation (the curve shape tracks) and a tight, roughly constant multiplicative offset — the device additionally applies an adjustment factor, an insulin divisor, and an autosensitivity ratio, none of which the replay models — rather than a ratio of exactly one.

This step also surfaced a data-quality issue: four users switched the *units* of their logged ISF mid-history (mmol/L per unit vs mg/dL per unit). A per-reading correction (values below 20 scaled by 18.018) resolves all four. It affects only this validation read; the main pipeline never consumes that field.

5. How the outputs support the conclusions

v2 is the gentler equation for almost everyone. Across the replayed cohort, v2 gives a higher ISF, and so a weaker correction, than v1 on 92% of readings, a median of $3.0\times$. The margin is widest at low glucose, where the v2 log nears its floor, and narrows to about $1.5\times$ when glucose is high. Both differences between the equations push the same way at realistic doses: the steeper TDD law and the altered glucose term each lift v2's ISF above v1's, so at target glucose the two would only meet near 194 U/day, well beyond anyone in the cohort. The pattern is a direct reading of the replayed data, not a modelling artefact.

Both TDD power laws are steeper than the fitted proxy relationship. The log-log fit of sensitivity against reconstructed TDD has a slope near -0.5 (bootstrap interval excluding -1). v1 assumes -1 and v2 assumes -2 . The proxy is calculated from the same closed-loop record, so this comparison does not establish a physiological TDD exponent.

A square-root law fits these two targets best. Under cross-validation the $K/\sqrt{\text{TDD}}$ form is best against tuned-profile ISF and tied-best against the delta-IOB proxy, beating both equations and the other candidates tested. Added inputs (carb ratio, target, basal fraction) do not robustly improve it. This is a predictive ranking against two observational targets. It does not settle the physiological shape or level.

The implementation is faithful. The unit tests fix the implementation to the defined equations, and the device-ISF validation shows the dynamic-ISF users tracking their devices' logged ISF with stable per-person offsets once the units issue is corrected, with no unexplained outliers. This licenses treating the replayed ISF as the equation's true output.

6. Reproducibility

Step	Script	Key output
Equation implementation + tests	inv008/dynisf.py, inv008/tests/	27 passing tests
TDD reconstruction	inv008/stage1_tdd.py (+ tdd_windows.py, sources.py)	per-person TDD tables
ISF replay	inv008/stage2_replay.py	per-person ISF tables
Orchestration	inv008/runner.py	run logs + manifests
Figures	inv008/stage3_plots.py	cohort + per-person figures
Equation search	fit_best_isf.py	candidate-comparison tables
Historical personalisation fit	inv008/ fit_personalisation.py	proxy exponent / per- user-K description; dosing interpretation withdrawn
Device validation	inv008/ validate_device_isf.py	validation tables
Delivery/basal inputs	extract_treatments_tdd.py, extract_hourly_basal.py	input data
Reference cohort + delta-IOB proxy	canonical_cohort.py, empirical_isf_v5.py	targets for the search
python -m pytest inv008/tests/ python -m inv008.runner --stage 1 --platforms v6 v7 python -m inv008.runner --stage 2 --platforms v5 v6 v7 python -m inv008.stage3_plots python fit_best_isf.py python -m inv008.validate_device_isf		

7. Limitations

1. **Counterfactual replay.** We compare the ISF each equation *would have computed* on real histories, not closed-loop glycaemic outcomes.
2. **Basal approximation for one platform.** Where exports lack temp-basal records, basal TDD uses the profile schedule.

3. **Delta-IOB proxy.** The regression is affected by in-window delivery, unrecorded carbohydrate, endogenous-glucose effects and closed-loop confounding. It has confidence intervals but no external physiological ground truth. Its level and TDD slope are provisional.
4. **Single cohort.** Open-source AID users, mostly 2016–2023; no commercial-system or demographic data. $n = 114/138$ for the historical proxy analyses.
5. **Anchor uncertainty.** Five reconstructed users retain time-anchor uncertainty and are flagged.